

DILIPKUMAR NEHRU

Backend Engineer — Node.js | TypeScript | MongoDB | SaaS Platforms

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PROFESSIONAL SUMMARY

Backend Engineer with 4+ years building and scaling SaaS platform APIs using Node.js, TypeScript, Express.js, and NestJS. Experienced across the full backend lifecycle — schema design and query optimisation on MongoDB, asynchronous job/queue processing with RabbitMQ, JWT/OAuth2 authentication and role-based access control, and API documentation. Delivered production services handling 50,000+ daily requests at 99.5% uptime for enterprise CRM and analytics platforms, including webhook delivery, real-time event pipelines, and multi-tenant SaaS architecture. Comfortable owning services end-to-end: design, implementation, testing, monitoring, and production support.

CAREER HIGHLIGHTS

- 50,000+ daily API requests served with 99.5% uptime across enterprise CRM and analytics platforms
- 35% reduction in API latency via MongoDB indexing, query optimisation, and connection pooling
- Job/queue processing time cut from 8s to under 2s using RabbitMQ consumer redesign and dead-letter queues
- 40% faster regression cycles through automated test suites (Selenium, Postman, JMeter)

WORK EXPERIENCE

Backend Engineer — Software Developer | Triotech Bizserve Private Limited

Jun 2026 – Sep 2026

Ringingo — multi-tenant cloud telephony SaaS platform

- Developed and enhanced backend services for a multi-tenant SaaS platform supporting real-time CDR processing, webhook delivery, and third-party API integrations.
- Designed a RabbitMQ-driven real-time processing pipeline with timezone-aware scheduled webhooks, incremental cursor-based delivery, and retry/delivery-log mechanisms — directly applicable to async workflow and job-queue requirements.
- Built API-key and JWT-based authentication (x-api-key and Bearer token support) with key refresh and expiry handling, supporting secure API access patterns.
- Delivered end-to-end features across NestJS/Node.js with TypeScript, MongoDB, microservices, and React/Next.js in a multi-tenant, scalable SaaS environment.

Software Engineer — Backend Developer | Zuqo Corporation India Pvt Ltd

Dec 2023 – May 2025

- Designed and maintained scalable RESTful APIs using Node.js and Express.js, serving 50,000+ daily requests with 99.5% uptime for enterprise web applications.
- Reduced MongoDB query response times by 35% through strategic indexing, query optimisation, and connection pooling on a healthcare CRM platform.
- Implemented asynchronous background task processing with RabbitMQ, cutting average job completion time from 8s to under 2s for an AI analytics platform.
- Standardised MongoDB and MySQL schema design patterns across backend microservices, reducing inter-service data inconsistencies by 60%.
- Implemented JWT-based authentication and OAuth 2.0 authorisation flows across multiple APIs, eliminating session vulnerabilities and enabling secure third-party integrations.
- Integrated Socket.io for real-time WebSocket notifications and live dashboard updates, reducing client-side polling overhead by 80%.
- Debugged and resolved 30+ production incidents, reducing Mean Time to Resolution (MTTR) by ~45%; collaborated in 2-week Agile sprints with zero critical production rollbacks.
- Enforced clean code standards and GitFlow code-review practices, reducing post-merge defect rate by 25%.

Software Test Engineer — QA Automation | Zuqo Corporation India Pvt Ltd

Apr 2022 – Dec 2023

- Built and maintained a Selenium automation suite covering 200+ functional and regression test cases, cutting manual testing effort by 40%.

- Performed API testing and backend validation using Postman and JMeter, identifying 80+ defects prior to production deployment — foundation for unit/integration test discipline.
- Created structured bug reports, test plans, and troubleshooting runbooks adopted as the team's QA documentation standard.

PROJECT HIGHLIGHTS

NovoNordisk CRM — Healthcare Enterprise Platform

- Architected scalable backend APIs (Node.js, Express.js, MongoDB) handling 50,000+ daily requests for a global pharmaceutical CRM.
- Implemented JWT authentication, OAuth 2.0, field-level encryption, and RBAC for secure handling of sensitive patient and sales data — directly relevant to GDPR-aligned data-handling practices.
- Optimised 15 high-frequency MongoDB aggregation queries, reducing dashboard load times by 35% for 500+ concurrent users.

UrbanPiper AI Analytics — F&B Analytics Dashboard

- Built event-driven backend microservices with RabbitMQ and Socket.io to stream real-time analytics to 100+ restaurant client dashboards.
- Reduced async job processing latency from 8s to under 2s by redesigning queue consumers and adding dead-letter queue error handling.
- Delivered REST APIs consumed by React dashboards with sub-500ms response times.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, SQL, Python, Java

Backend Frameworks: Node.js, Express.js, NestJS, Socket.io

Databases & ORMs: MongoDB (schema design, indexing, aggregation), Mongoose, MySQL, PostgreSQL

Messaging & Async: RabbitMQ, job/queue processing, event-driven architecture, WebSockets, webhook/retry mechanisms

Auth & Security: JWT, OAuth 2.0, role-based access control (RBAC), bcrypt, HTTPS/TLS, field-level encryption

Frontend: React.js, Next.js, Tailwind CSS

Testing: Postman, Selenium WebDriver, JMeter, API automation, unit & integration testing practices

DevOps / Tools: Git, GitHub, Bitbucket, Linux CLI, Agile/CI-CD awareness

API Design: RESTful API design, API versioning, microservices, multi-tenant SaaS architecture

CERTIFICATIONS

- Selenium Automation & Manual Testing — QSpiders, Bangalore (2022)
- MongoDB Basics (M001) — MongoDB University (in progress)
- Node.js Application Developer — OpenJS Foundation (in progress)

EDUCATION

Bachelor of Engineering – Electronics & Communication Engineering

RVS Technical Campus, Coimbatore (Anna University) — 2020

Diploma – Electronics & Communication Engineering

Krishnasamy Memorial Polytechnic College, Cuddalore — 2012 – 2015